12.287

AI25BTECH11027 - NAGA BHUVANA

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Question:

An Unitary matrix $\begin{pmatrix} ae^{i\alpha} & b \\ ce^{i\beta} & d \end{pmatrix}$ is given where $a, b, c, d, \alpha, \beta$ are real. The inverse of the matrix is

a) $\begin{pmatrix} ae^{i\alpha} & -ce^{i\beta} \\ b & d \end{pmatrix}$ b) $\begin{pmatrix} ae^{i\alpha} & ce^{i\beta} \\ b & d \end{pmatrix}$ c) $\begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix}$ d) $\begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix}$

Solution:

Unitary Matrix: Complex square matrix whose inverse is equal to its conjugate transpose

Let

$$M = \begin{pmatrix} ae^{i\alpha} & b \\ ce^{i\beta} & d \end{pmatrix} \tag{0.1}$$

$\bar{M}$ be Complex Conjugate Matrix of M

$$\bar{M} = \begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix} \tag{0.2}$$

$$M^{-1} = \bar{M}^T \tag{0.3}$$

$$M^{-1} = \begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix}^T \tag{0.4}$$

$$M^{-1} = \begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix} \tag{0.5}$$

$\therefore$ Inverse of the Matrix is $\begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix}$